Southern New Hampshire University

DriverPass Business Requirements Document

CS 255 Project One

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Document purpose. This document defines the business and system requirements for DriverPass, a driver-training company that will provide online learning, practice tests, and scheduled two-hour driving lessons. The recommended solution is a secure, cloud-hosted web system that centralizes customer, employee, appointment, package, training, and audit information.

System Components and Design

Purpose

- The client is DriverPass, represented by owner Liam and IT officer Ian.
- The project will create a centralized system for online driver education, practice tests, customer registration, training-package management, and on-the-road lesson scheduling.
- The system will let customers and employees access current information from computers and mobile devices while reducing manual scheduling, duplicate records, and uncertainty about appointments.
- The system will support management reporting, employee accountability, customer communication, and updates from the DMV.

System Background

- DriverPass is responding to the high rate of DMV driving-test failures by offering structured education and practical training.
- Customers need online classes and practice tests, optional in-person instruction, and two-hour driving sessions with assigned vehicles and drivers.
- The current business concept relies on phone or office registration and appointment scheduling, which can create conflicts and makes changes harder to track.
- The proposed system needs a customer-facing website, role-based employee portal, cloud database, scheduling component, online course and testing component, reporting tools, notification capability, and secure integration with DMV updates.
- Users include customers or students, the secretary, driving instructors, the owner, and the IT administrator. Each role requires different permissions.

Objectives and Goals

- Provide secure account registration, login, password reset, and role-based access for every user type.
- Allow customers or the secretary to create, modify, and cancel two-hour lesson reservations without assigning the same driver or vehicle to overlapping appointments.
- Match every scheduled lesson to one customer, one available driver, one available vehicle, a pickup location, and the same drop-off location.
- Offer the three specified training packages and allow authorized staff to enable or disable packages without a code change.
- Provide online course content and practice tests, with test name, elapsed time, score, and status recorded for each attempt.

- Record instructor lesson times and comments so staff and customers can review training progress.
- Receive DMV rule, policy, and sample-question updates and notify authorized staff when new material is available.
- Record who creates, changes, or cancels reservations and produce downloadable activity and business reports.
- Complete the planned project from requirements collection through client sign-off according to the approved schedule.

Requirements

Nonfunctional Requirements

Performance Requirements

- **Web availability:** The system shall operate as a responsive web application available from current desktop and mobile browsers.
- **Response time:** The system should return normal page and scheduling results within three seconds under expected business load.
- **Concurrent use:** The system shall support customers and employees using scheduling, testing, and reporting functions at the same time without data conflicts.
- **Data currency:** Reservation, driver, vehicle, package, and test-progress changes shall be visible immediately after a successful transaction.
- **Content updates:** Authorized staff shall be able to publish DMV and training-content changes without taking the entire system offline.
- **Availability and recovery:** The cloud provider shall perform automated backups and support restoration after service or data failure.

Platform Constraints

- **Cloud hosting:** The system shall use a managed cloud environment so DriverPass does not maintain local servers, backups, or infrastructure security.
- **Browser compatibility:** The interface shall support current versions of major browsers on Windows, macOS, iOS, and Android.
- **Central database:** The back end shall use a centralized transactional database for users, roles, packages, appointments, vehicles, drivers, course content, tests, payments, and audit records.
- **Online updates:** Users may download reports for offline review, but system records shall only be changed while connected to the central service to prevent duplicate or conflicting data.
- **External connection:** The design shall provide a controlled integration point for DMV update notifications and content imports.

Accuracy and Precision

- **Unique identity:** Every user, vehicle, appointment, test attempt, and training package shall have a unique system identifier.
- **Credential handling:** Usernames shall be treated consistently, while passwords shall remain case-sensitive and never be stored as readable text.
- **Scheduling precision:** Appointment times shall be stored with dates, start times, end times, and the applicable time zone; each lesson shall last two hours.

- **Conflict validation:** The system shall reject overlapping assignments for the same customer, driver, or vehicle and display a specific correction message.
- **Input validation:** Required customer, payment, location, and scheduling fields shall be checked for missing or incorrectly formatted values before submission.
- **Administrative alerts:** The system shall notify the IT administrator of repeated failures, integration errors, backup failures, and suspected unauthorized access.

Adaptability

- **User management:** Authorized administrators shall add, disable, reactivate, and modify users and role assignments without changing code.
- **Package control:** Authorized business users shall enable or disable existing training packages; adding new modules or restructuring packages may require a future development release.
- **Configurable content:** Authorized staff shall update course content, practice questions, policies, contact information, and other website content through administrative screens.
- **Maintainability:** The application shall separate the user interface, business rules, and data-access logic so platform or browser updates can be handled with limited impact.
- **IT access:** The IT administrator shall have full account-management, configuration, monitoring, audit, and troubleshooting access, but payment data access shall be limited to business need.

Security

- **Authentication:** Every employee and registered customer shall use a unique account and password; privileged employee accounts should use multifactor authentication.
- **Encryption:** All browser-to-server communication shall use HTTPS, and sensitive stored data shall be encrypted using the cloud provider's supported controls.
- **Payment protection:** Card data shall be processed through a compliant payment service. The system should store a payment token and transaction reference rather than a security code or complete card number.
- **Authorization:** Role-based access shall limit customers, secretaries, drivers, the owner, and IT staff to the information and actions required for their work.
- **Brute-force protection:** Repeated failed login attempts shall trigger rate limiting, temporary account lockout, audit logging, and an administrator alert when thresholds are exceeded.
- **Password recovery:** Users shall reset forgotten passwords through a time-limited verification link or code; administrators may force employee resets and disable separated employees immediately.
- **Auditability:** Security-sensitive events and all reservation creation, modification, and cancellation actions shall record the user, date, time, and action performed.
- **Session protection:** The system shall expire inactive sessions and require users to authenticate again before sensitive account or payment changes.

Functional Requirements

- The system shall register customers and store their first name, last name, address, phone number, state, pickup location, and contact information.
- The system shall authenticate users, automatically support verified password resets, and enforce permissions by role.

- The system shall allow IT administrators to reset, disable, and restore accounts and assign access rights.
- The system shall display the three DriverPass packages and allow authorized staff to enable or disable package enrollment.
- The system shall enroll a customer in a selected package and track the remaining training hours and included services.
- The system shall allow customers and secretaries to create, modify, and cancel two-hour driving appointments.
- The system shall assign each appointment to one available driver and one available vehicle and prevent scheduling conflicts.
- The system shall require the lesson drop-off location to match the pickup location unless DriverPass later changes this policy.
- The system shall provide online course materials and practice tests to customers whose packages include them.
- The system shall record each test's name, time taken, score, and status as not taken, in progress, failed, or passed.
- The system shall allow instructors to record lesson start time, end time, and driver comments.
- The system shall show customers their appointments, test progress, package progress, and instructor comments appropriate for customer viewing.
- The system shall allow authorized staff to contact a student and shall provide customers with a contact page.
- The system shall receive or import DMV rule, policy, and sample-question updates and notify authorized staff of new updates.
- The system shall record the creator and most recent modifier of reservations and maintain a history of cancellations and changes.
- The system shall generate printable and downloadable activity, reservation, customer, training-progress, and business reports, including spreadsheet-compatible exports.
- The system shall process customer payments through a secure external payment service and record payment status without exposing sensitive card data.
- The system shall create automated backups and support recovery of system and customer data.

User Interface

- Customers will use a responsive browser interface to manage their accounts, select packages, schedule or change appointments, access course materials and practice tests, view progress, reset passwords, and contact DriverPass.
- The secretary will use an employee dashboard to register customers, maintain contact information, schedule appointments on a customer's behalf, and review driver and vehicle availability.
- Driving instructors will view assigned lessons and enter lesson times and comments.
- The owner will view operational dashboards, reports, package status, customer progress, and audit activity.
- The IT administrator will manage accounts, roles, system configuration, monitoring, integrations, and security events.

- Interfaces shall use consistent navigation, clear field labels, confirmation messages, specific error feedback, accessible color contrast, and warnings before destructive actions.
- The student progress page shall display test name, time taken, score, and status. The driver-notes area shall display lesson time, start hour, end hour, and driver comments.

Assumptions

- Customers and employees have internet access and use a reasonably current browser on a computer or mobile device.
- DriverPass will provide branding, final page designs, course content, practice questions, employee records, vehicle records, package pricing, and business policies before implementation milestones require them.
- The final cloud, payment, email or text notification, and DMV-integration vendors will be selected during detailed design.
- A car normally has a primary driver, but the database will maintain driver and vehicle availability separately so authorized staff can reassign resources when needed.
- All current on-the-road lessons are exactly two hours, and pickup and drop-off locations are the same.
- Downloaded reports are reference copies; the central online database remains the authoritative source.
- DMV updates will be available through an authorized feed, import, or administrator workflow that DriverPass is permitted to use.

Limitations

- The initial release will allow existing packages to be enabled or disabled but will not provide a no-code tool for creating arbitrary new course modules or package structures.
- Offline users may view downloaded information but cannot update central records until they reconnect.
- System availability depends on internet service and the selected cloud and third-party providers.
- DMV automation is limited by the format, reliability, authorization, and availability of the external DMV update source.
- The project schedule assumes timely client approval on March 10 and 11 and no major change requests after approval.
- The transcript does not provide a final budget, expected number of users, formal uptime target, data-retention period, or legal jurisdiction; these items require client confirmation before implementation.
- The initial scope focuses on the stated training, scheduling, testing, reporting, and administrative needs; future features require separate prioritization and approval.

Gantt Chart

The schedule below follows the dates agreed upon by Sam and Jennifer. Modeling and interface-research tasks overlap where no dependency exists. Customer approval is the gate for interface construction, followed by database integration, business logic, testing, delivery, and sign-off.

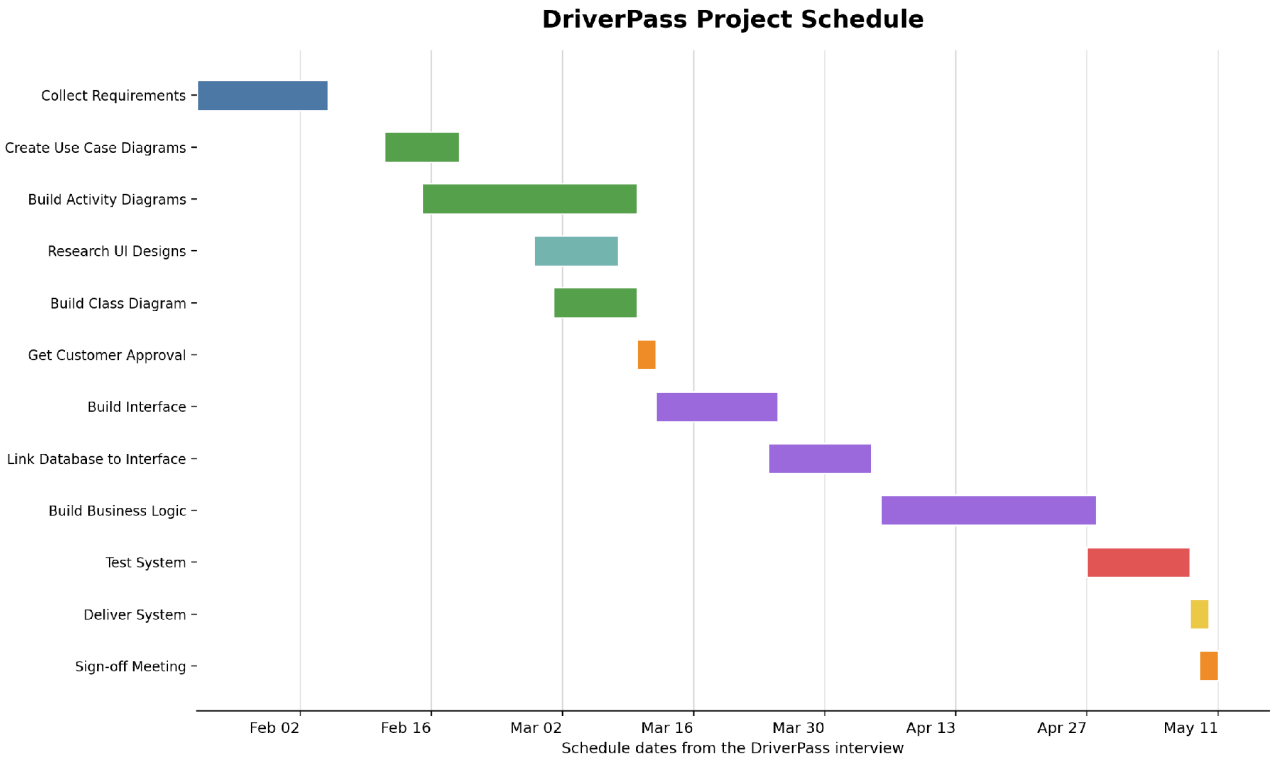


Figure 1 DriverPass project schedule based on the interview transcript